7408

[This question paper contains of ...... printed pages.]

Sr. No. of Question Paper: 7408

18.06.2024(M)

Your Roll No....................

Unique Paper Code:	**12271202 OC**
Name of the Paper:	**Mathematical Methods for Economics-II**
Name of the Course:	**B.A.(Hons.) Economics**
Semester:	**II**

Duration: 3 Hours

Maximum Marks: 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. There are 3 questions in all.
3. All questions are compulsory.
4. All parts of a question must be answered together.
5. Use of simple calculator is allowed.

1. Answer any *four*:

4 x 5 = 20

(a) Find the equation of the plane in R^3 through the point $(3,2,-4)$ with $p = (-2,4,3)$ as the normal.

(b) Solve for x_1, x_2 and x_3 given

$$3x_1 + 4x_2 - 2x_3 = -5$$
$$2x_1 + 3x_3 = 6$$
$$6x_1 + 5x_3 = -10$$

(c) Compute $(P - 2Q)$ when $P = \begin{pmatrix} 11 & 20 & 3 \\ 0 & -49 & -8 \end{pmatrix}$ and $Q = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$.

(d) Determine the rank of the following matrix:

$$B = \begin{bmatrix} -2 & 0 & 3 \\ 11 & 1 & 12 \\ 2 & -5 & 2 \end{bmatrix}$$

(e) Let $a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$, $c = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$. Check whether the vectors are linearly independent.

2. Answer any *four* of the following: $4 \times 5 = 20$

a) Given Cobb-Douglas production function. $Y = 10x^2y3$, find the elasticity of substitution.

b) Given $Y = AL^2K^3$, where Y is the aggregate output of the economy, L is labour force and K is the capital stock. If the capital stock and the labor force are growing according to $K(t) = K_0e^{2t}$ and $L(t) = L_0e^{4t}$

(i) What are the proportional growth rates of labor and capital?

(ii) What is the rate of growth of output?

c) Find the equation of the tangent plane to the surface $z = x^2 + y^2 + xy$ at the point $(1, -1, 1)$.

d) Examine the definiteness of the following quadratic forms:

(i) $f(x, y) = -x^2 + 4xy + 2y^2$

(ii) $3x^2 - y^2$ subject to $x - 2y = 7$

e) Find and sketch the domain of the following functions:

i) $f(x, y) = \sqrt{x^2 + y^2} + \sqrt{y}$

ii) $f(x, y) = \sqrt{9 - (x^2 + y^2)}$

[For PWD candidates in lieu of part e]

Find the domain of the following functions:

i) $f(x, y) = \sqrt{x^2 + y^2} + \sqrt{y}$

ii) $f(x, y) = \sqrt{9 - (x^2 + y^2)}$

iii) $f(x, y) = \sqrt{x^2 - y}$

3. Answer any five of the following: $7 \times 5 = 35$

(a) Show that any function $x = x(t)$ that satisfies the equation $(t - a)^2 + x^2 = a^2$ is a solution of the following differential equation:

$$2tx\frac{dx}{dt} + t^2 - x^2 = 0$$

(b) Find quadratic approximation for the function: $z = e^{x+2y}$ at the point $x = 0, y = 0$.

(c) Find all extreme points and/or saddle points of the function:

$$f(x,y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$$

Verify the second order conditions.

(d) Find the maximum value of the utility function: $U(x,y) = 100 - e^{-x} - e^{-y}$ subject to $px + qy = m$, using Lagrange multiplier method. Verify the second order conditions.

(e) Find the global extreme values of $f(x,y) = 2x - 3y$ defined over the set $\{(x,y): x^2 + y^2 \leq 16\}$.

(f) Given differential equation $\dot{x} = x + t$, show that $x = -t - 1$ is a particular solution.